

MISINFORMATION DETECTOR

for Immigration News in Spanish Digital Media

Milestone 2. Model Building and Evaluation

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1. Introduction

Misinformation in digital media represents a significant challenge for public understanding and policy, particularly in Spain. Social media platforms, online news outlets, and public communication channels can rapidly propagate false or misleading content, impacting political, social, and health-related discourse.

This project addresses the detection of misinformation by proposing an end-to-end data processing pipeline capable of ingesting content, classifying claims, retrieving supporting evidence, and providing explainable feedback through a chatbot interface. The focus is on Spanish political news, particularly migration-related content. The primary objective is to detect misinformation early and equip users with tools to verify, understand, and counter false claims.

2. Data Preparation

2.1 Dataset Construction

A custom dataset was created by collecting news from multiple verified Spanish sources (Cadena SER (n.d.), El País (n.d.), Agencia EFE (n.d.), Maldita Migración (n.d.)). All sources were unified into a common schema of eight columns: title, label, text, topic, URL, publication date, medium, and extraction date. The initial real-only dataset contained news items labelled TRUE, FALSE, or CONTEXT. Eleven news items categorized as ALERT were reclassified as TRUE, as they were verified alerts.

We generated a synthetic dataset of manipulated news items from the real ones. The synthetic schema comprises seven columns: title, label, text, topic, technic, modification, and publication date. Synthetic items were labelled as either FALSE or CONTEXT and then integrated with the real news.

2.2 Analysis of Label Distribution

Figure 1 shows a grouped bar chart comparing the real-only distribution against the final combined dataset. The real data exhibits a severe imbalance: TRUE dominates, while FALSE and CONTEXT are negligible. After adding synthetic data, the counts for FALSE and CONTEXT rise substantially, while TRUE remains unchanged. The final combined dataset reduces the original 11:1 ratio to approximately 0.86:1, achieving a practically balanced corpus.

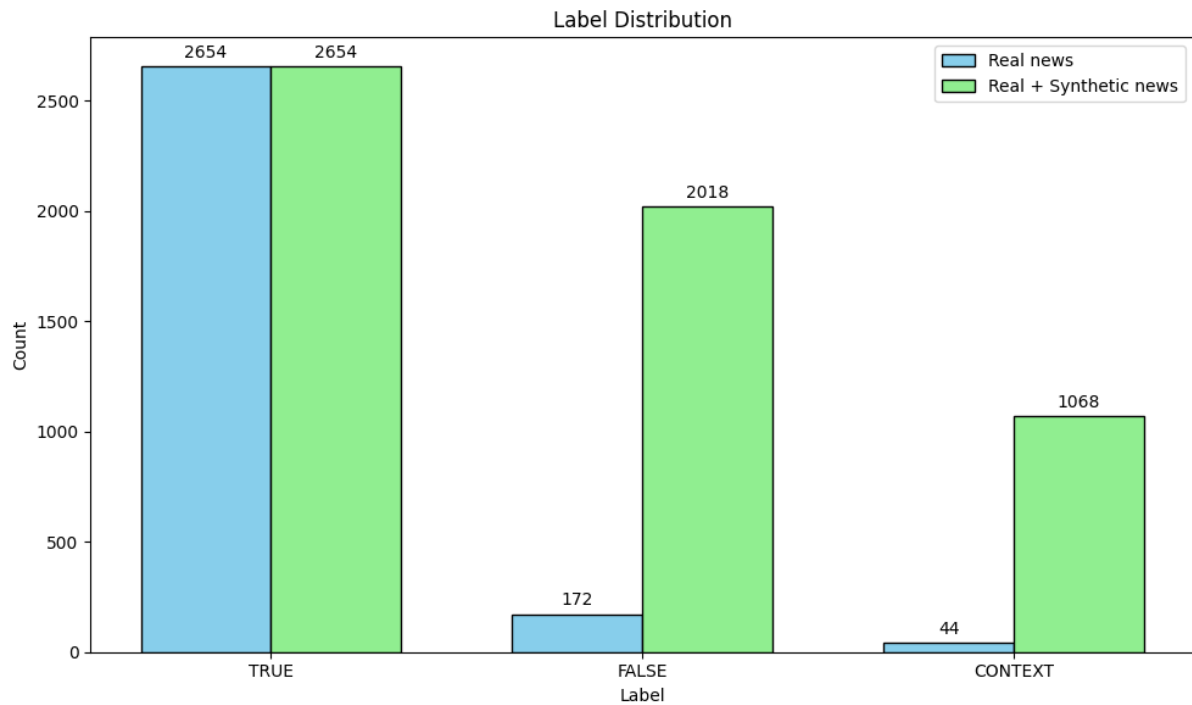


Figure 1: Label distribution of real news versus combined real and synthetic news

2.3 Cleaning and Transformation Pipeline

The preparation pipeline applied the following transformations to the news dataset:

- Null removal: Rows with a null title or text were eliminated using `dropna()`.
- Length filtering: Entries with a title shorter than 5 characters or a text shorter than 20 characters were discarded.
- Normalisation: Whitespace was stripped and UTF-8 encoding was standardised.
- Stratified split: The data was partitioned into 80% training, 10% validation, and 10% test sets using `train_test_split` with `stratify=label`, thereby preserving the class proportions in each partition.

2.4 Data Understanding and Exploratory Data Analysis

Text length analysis of the news revealed a mean of approximately 1200 characters per item (title+text), with the 90th percentile at around 3000 characters, equivalent to 450-500 XLM-RoBERTa tokens. Concatenating the title and text into a single model input maximises the information available for classification. This concatenated input remains within the 512-token limit of XLM-RoBERTa for the 85th percentile of examples.

To fully comprehend the nature of the misinformation in our statistics dataset, we conducted an Exploratory Data Analysis. We analyzed the main thematic pillars that drive migration-related news. We focused on three main themes for this analysis:

1. Arrivals

A big part of the fake news in our dataset is about how many migrants arrive in Spain and which countries they come from. To understand the real situation, we created a world map using colors to show the origins of migrants in 2024 (Illustration 1). This map clearly shows the main source countries. As we can see, there is a strong movement of people from Latin America (especially Colombia and Venezuela) and North Africa (Morocco).

2. Criminality

Task description

The task is a classification problem. Given a news item about immigration, the system must classify it into one of the categories: TRUE, FALSE, CONTEXT. The input is the text of the news item; the output is a label, a confidence score, and an alert level.

The model is designed to work as part of a larger system:

- The BETO model analyses the text and returns a label (TRUE or SUSPICIOUS/FALSE), a probability, and an alert level (VERY LOW / LOW / MEDIUM / HIGH).
- The RAG system searches for similar verified news items as evidence.
- The chatbot's LLM uses the model's output and the RAG evidence to generate an explained response in natural language.

The model's sole function is to provide the classification and confidence. It does not explain why a news item is false, that is done by the LLM with RAG evidence.

Materialising minable view

In this project, the problem is formulated as a supervised text classification task.

- Input (X): The text of a news item, represented as a string.
- Output (Y): A categorical label indicating whether the news is TRUE, FALSE, CONTEXT, or ALERT.
- Sample unit: One individual news item.
- Dataset size: The final dataset consists of 5,740 news items after balancing.

In terms of feature representation, two approaches are considered. A baseline model uses TF-IDF vectors to transform text into numerical features. The final model uses BETO embeddings, which capture deeper semantic and contextual information in Spanish text, improving classification performance.

4. Model building

4.1 Problem Formulation

The task is formulated as a multiclass text classification problem, where each input instance (news headline + body) must be assigned to one of three categories: VERDADERO (verified information), FALSO (misinformation), or CONTEXTO (partially true but misleading). This goes beyond binary fake news detection and better reflects real-world misinformation dynamics.

4.2 Dataset Preparation

The final dataset combines 2,870 real scraped news articles (labelled VERDADERO/FALSE/CONTEXTO) with 2,870 AI-generated distorted versions (labelled FALSO or CONTEXTO), totalling 5,740 instances. Combining both sources was necessary because the real dataset alone contains very few FALSO and CONTEXTO examples, making balanced three-class training difficult. After cleaning, the dataset is split 80/10/10% (train/validation/test) using stratified sampling to preserve class proportions.

4.3 Model Selection and Architecture

We fine-tune two Transformer-based models: **BETO** (dccuchile/bert-base-spanish-wwm-cased), pretrained exclusively on Spanish, and **XLm-RoBERTa** (xlm-roberta-base), a multilingual model. This allows us to compare domain specialisation versus generalisation capacity. Traditional approaches such as TF-IDF with logistic regression or SVM were discarded because they fail to capture the semantic nuance required to distinguish CONTEXTO from VERDADERO — a distinction that depends on subtle

framing and context rather than keyword frequency. Both models share the same architecture: a pretrained Transformer encoder, a fully connected classification head, and a softmax output over 3 classes. The input is the concatenated title and body, tokenized and truncated to 512 tokens.

4.4 Training Configuration

Both models are trained with AdamW, cosine LR scheduler with 10% warmup, early stopping (patience = 3), and FP16 mixed precision. Key hyperparameters are summarised below:

Hyperparameter	BETO	XLM-RoBERTa
Learning rate	2e-5	1.5e-5
Batch size (per GPU)	16	32
Max epochs	6	6
Best model metric	F1_falso	F1_macro

The learning rate and batch size differences are justified by model size and available hardware respectively. To handle class imbalance, both models apply inverse-frequency class weights and label smoothing ($\alpha = 0.1$), which reduces overconfidence and improves probability calibration. A confidence threshold of 0.60 flags low-confidence predictions as INCIERTO rather than forcing an incorrect classification. Different evaluation metrics were used during training. BETO uses F1 for the FALSE class, while RoBERTa uses macro F1. This results in different optimization behaviors and is treated as part of the experimental comparison.

4.5 Integration and Experimental Strategy

The classifier is one component of a larger pipeline: it produces a label, confidence score, and a prompt_hint that is passed to an LLM, which combines this verdict with documents retrieved by a RAG system to generate a full forensic explanation for the user. Both models are fully trained and evaluated under identical conditions; XLM-RoBERTa is currently deployed in the chatbot while final model selection will be determined by the evaluation results in Section 5.

5. Evaluation

5.1 Evaluation Metric Justification

Given the multiclass nature of the task and the class imbalance present in the dataset, accuracy alone is insufficient as an evaluation metric. We use macro F1-scores as the primary metric, which treats all classes equally regardless of their frequency. We also report per-class precision, recall and F1, as well as weighted F1 and overall accuracy. This combinations allows us to asses both global performance and the model's behaviour on each individual class, particularly the minority class CONTEXTO.

5.2 Results

Both models were evaluated on the same held-out test set (573 instances, stratified). Results are summarised below:

Metric	BETO	XML-RoBERTa
Accuracy	0.838	0.846
F1 Macro	0.749	0.782
F1 Weighted	0.824	0.843
F1 VERDADERO	0.991	0.994
F1 CONTEXTO	0.455	0.554
F1 FALSO	0.801	0.798

Per-class results – BETO:

Class	Precision	Recall	F1	Support
VERDADERO	0.992	0.989	0.991	264
CONTEXTO	0.633	0.355	0.455	107
FALSO	0.742	0.896	0.801	202

Per-class results – XML-RoBERTa:

Class	Precision	Recall	F1	Support
VERDADERO	1.000	0.989	0.994	264
CONTEXTO	0.614	0.505	0.554	107
FALSO	0.759	0.842	0.798	202

5.3 Training Behaviour

BETO converged in 4 epochs (early stopping triggered), achieving its best validation F1_falso of 0.801 at epoch 1.

Epoch	Training Loss	Validation Loss	Accuracy	F1 Macro	F1 Weighted	F1 Falso
1	0.672436	0.662023	0.825480	0.716814	0.805232	0.785249
2	0.619711	0.622900	0.808028	0.762769	0.810799	0.666667
3	0.547921	0.634254	0.834206	0.782159	0.836961	0.758974
4	0.443641	0.735547	0.837696	0.779897	0.838085	0.773399

XML-RoBERTa ran all 6 epochs, reaching its best macro F1 of 0.783 at epoch 4.

Epoch	Training Loss	Validation Loss	Accuracy	F1 Macro	F1 Weighted	F1 False
1	0.978531	0.741924	0.794066	0.677988	0.766136	0.753304
2	0.691416	0.640365	0.818499	0.771065	0.823403	0.714681
3	0.606289	0.642274	0.792321	0.742262	0.796972	0.658824
4	0.555036	0.645879	0.849913	0.783438	0.844633	0.800926
5	0.539183	0.659617	0.834206	0.776359	0.834578	0.767327
6	0.517412	0.676319	0.835951	0.776276	0.835409	0.772616

Both models show a steady decrease in training loss, with validation loss stabilising after epoch 2, indicating no significant overfitting.

5.4 Analysis

Both models perform strongly on VERDADERO ($F1 > 0.99$) and achieve solid results on FALSO ($F1 \approx 0.80$), the most critical class for misinformation detection. The most challenging class is CONTEXTO (BETO: 0.455, XLM-RoBERTa: 0.554), which is expected given its smaller size, its linguistic similarity to VERDADERO, and the inherent ambiguity of the class itself. XLM-RoBERTa outperforms BETO on CONTEXTO by +0.10 F1 and achieves higher macro F1 overall (0.782 vs 0.749), suggesting its multilingual pretraining generalises better to subtle semantic distortions. Based on these results, XLM-RoBERTa is selected for deployment in the final pipeline.

5.5 Limitations

Both models show low confidence on short headline-only inputs (all predictions below the 0.60 threshold in the inference test), confirming that the models require full article text to produce reliable predictions. This is consistent with the training setup, where both title and body were always provided as input.

Use of technology

The project makes use of several tools and technologies for data processing, model training, and deployment.

- Python was used as the main programming language.
- pandas was used for data cleaning, preprocessing, and dataset manipulation.
- scikit-learn was used for TF-IDF vectorisation and baseline models.
- Hugging Face Transformers was used to fine-tune the BETO model.
- PyTorch was used as the deep learning framework for training and inference.
- Google Colab with GPU (NVIDIA T4) was used for model training.

To ensure reproducibility, experiments were conducted in a controlled environment (Google Colab) with fixed training configurations (epochs, class weights, and dataset splits). The use of well-established libraries such as Hugging Face and PyTorch also facilitates reproducibility and future extensions of the project.

The team worked autonomously to configure the training environment, handle model fine-tuning, and integrate different components of the system, solving technical challenges during development.

References

1. Cadena SER. (n.d.). *Cadena SER*. <https://cadenaser.com>
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3. Agencia EFE. (n.d.). *Agencia EFE*. <https://efe.com>
4. Maldita.es. (n.d.). *Maldita Migración*. <https://maldita.es/migracion/>

Anexos

Migraciones por país en 2024

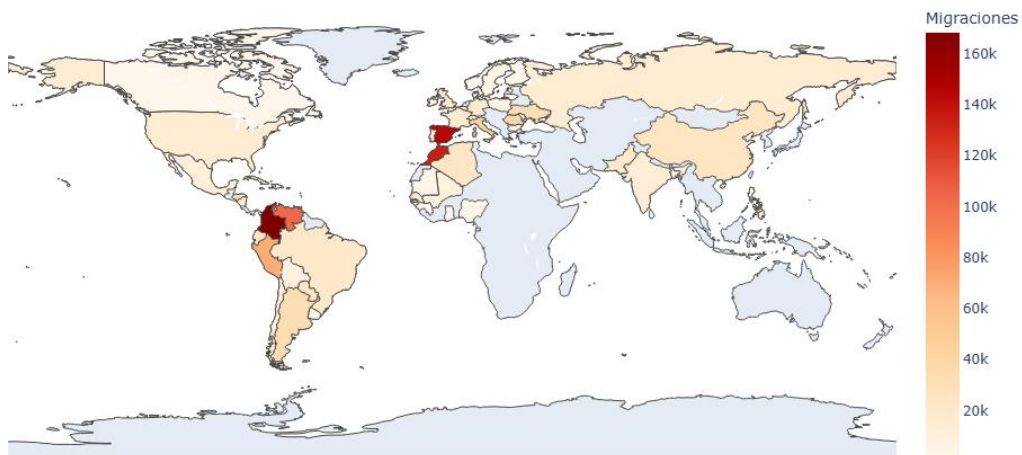


Illustration 1: Geographic distribution of migratory flows to Spain by country of origin (2024).

